SMART DISCLOSURE: DETERMINISTIC LAYER 0 HARDWARE INTERDICTION & STATE-SPACE BOUNDING FOR EDGE AI AND CRITICAL OT INFRASTRUCTURE

The Central Absolute Zero-Point Anchor & Licensing Tri-Barrier Specification

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Abstract

This technical specification establishes the formal **Layer 0 (L0) Smart Disclosure Anchor** for the 19-state bounded continuum $\mathcal{S} \in \{-9, \dots, \mathbf{0}, \dots, +9\}$. Positioned precisely at the **Absolute Zero-Point (Fulcrum** $= 0$**)**, Layer 0 represents the physical grounding of Data Physics, governing Direct-to-Chip (D2C) mCDU liquid cooling, high-voltage power transmission limits, and hardware interdiction. To complete the 19-node continuum $[L_{-9} \leftrightarrow L_0 \leftrightarrow L_9]$ and establish an unassailable defensive publication wall, this monograph incorporates a strict **Licensing Tri-Barrier**: (1) **GNU AGPLv3** for software/kernel logic, (2) **CERN Open Hardware Licence v2 - Strongly Reciprocal (CERN-OHL-S-v2)** for physical/mechanical schematics, and (3) **CC BY-NC-ND 4.0** for technical specifications and Prior Art disclosures.

1 Strategic Architecture: The L0 Fulcrum

Layer 0 is the physical and mathematical center of the entire architecture. While higher layers ($L_1 \dots L_9$) govern software, distributed clusters, and continental grids, and lower layers ($L_{-1} \dots L_{-9}$) govern quantum and pre-geometric physics, **Layer 0 measures raw physical thermodynamics**.

Defensive Prior Art Declaration & Patent Shield

This document constitutes an official public, time-stamped defensive disclosure under 35 U.S.C. §102 and EPC Art. 54. All concepts, state-space bounds, hardware interdiction triggers, and 19-state lattice definitions contained herein reference and extend Polish Patent Office (UPRP) filings **P.454742, P.457301, P.457157, P.457109**. Third-party patenting of deterministic Layer 0 state-space bounding or hardware interdiction is strictly precluded.

2 The Licensing Tri-Barrier

To protect open scientific infrastructure while enforcing strong reciprocity across commercial hardware implementations, Layer 0 operates under three distinct, non-overlapping licenses:

Domain	License Type	Enforcement Mechanism
Software & C99 RTOS Engine	GNU AGPL v3.0 Only	Copyleft network service trigger
Hardware, PCB & mCDU Schematics	CERN OHL v2 - Strongly Reciprocal	Reciprocal hardware modification derivat
Monograph, Prior Art & Standards	CC BY-NC-ND 4.0 International	Attribution, Non-Commercial, No Deriva

Table 1: The Layer 0 Licensing Tri-Barrier Architecture.

3 Formalism and C99 Execution Governor

The Layer 0 state governor executes in $O(1)$ constant time (< 0.35 ms), sampling physical sensors (thermal, pressure, voltage) and mapping them to $\mathcal{S} \in \{-9, \ldots, \mathbf{0}, \ldots, +9\}$.

```c
// Zero-Heap C99 Layer 0 State Governor
#include <stdint.h>
#include <signal.h>

#define HARD_STOP_INTERDICTION -999

typedef struct {
    int8_t thermal_state;   // [-9, +9]
    int8_t pressure_state;  // [-9, +9]
    int8_t grid_state;      // [-9, +9]
} L0_PhysicalVector;

int32_t audit_L0_fulcrum(const L0_PhysicalVector* phys) {
    if (phys->thermal_state >= 9 || phys->pressure_state >= 9 || phys->
        grid_state >= 9) {
        // Trigger OS-level SIGKILL (kill -9 -> POSIX exit status 137)
        raise(SIGKILL);
        return HARD_STOP_INTERDICTION;
    }
    return 0; // Nominal Zero-Point equilibrium
}
```

4 Conclusion

By establishing **L0 Smart Disclosure**, the 19-node continuum $[L_{-9} \leftrightarrow L_0 \leftrightarrow L_9]$ achieves complete structural and legal closure. Layer 0 remains the unyielding, deterministic anchor against stochastic AI infrastructure failures.